

TP DEVOPS : DOCKER – Marc Wanduragala

Objectifs :

1. Création des pages web

Créer trois pages PHP :

- index.php : Page d'accueil avec un message statique de bienvenue.
- about.php : Page À propos avec un texte statique sur l'application.
- contact.php : Page Contact affichant dynamiquement un message stocké dans la base de données (par exemple : "Contactez-nous au 0976436742").

Configurez un serveur Apache/PHP pour servir ces pages dans un conteneur.

2. Configuration Docker Compose

Écrivez un fichier docker-compose.yml pour orchestrer les deux conteneurs (Web et Base de données). Assurez-vous que :

- Les conteneurs sont interconnectés via le même réseau.
- Les ports externes sont mappés correctement pour accéder à l'application.
- Configurez un volume pour persister les données de MySQL.
- Montez les fichiers PHP via un volume dans le conteneur Apache pour faciliter les modifications.

:

Générer index.php :

```
marc@marc-virtual-machine:~$ vim index.php
```

Contenu de index.

```
1  <?php
2  // This is a simple PHP script for demonstration
3  $welcomeMessage = "Welcome to My Website!";
4  ?>
5
6  <!DOCTYPE html>
7  <html lang="en">
8  <head>
9      <meta charset="UTF-8">
10     <meta name="viewport" content="width=device-width, initial-scale=1.0">
11     <title>Basic Website</title>
12     <style>
13         body {
14             font-family: Arial, sans-serif;
15             background-color: #f4f4f9;
16             margin: 0;
17             padding: 0;
18             display: flex;
19             justify-content: center;
20             align-items: center;
21             height: 100vh;
22         }
23         .container {
24             text-align: center;
25             background: #fff;
26             padding: 20px;
27             border-radius: 8px;
28             box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
29         }
30         h1 {
31             color: #333;
32         }
33         p {
34             color: #666;
35         }
36     </style>
37 </head>
38 <body>
39     <div class="container">
40         <h1><?php echo $welcomeMessage; ?></h1>
41         <p>This is a simple PHP-powered web page.</p>
42     </div>
43 </body>
44 </html>
45 S
46
```

Générer about.php

Vim about.php

Contenu de about.php

```
1  <?php
2  // This is an About page for the website
3  $pageTitle = "About Us";
4  $aboutContent = "This website is a simple demonstration of PHP and HTML integration. We aim to help beginners learn the basics of web development.";
5  ?>
6
7  <!DOCTYPE html>
8  <html lang="en">
9  <head>
10     <meta charset="UTF-8">
11     <meta name="viewport" content="width=device-width, initial-scale=1.0">
12     <title><?php echo $pageTitle; ?></title>
13     <style>
14         body {
15             font-family: Arial, sans-serif;
16             background-color: #f4f4f9;
17             margin: 0;
18             padding: 0;
19             display: flex;
20             justify-content: center;
21             align-items: center;
22             height: 100vh;
23         }
24         .container {
25             text-align: center;
26             background: #fff;
27             padding: 20px;
28             border-radius: 8px;
29             box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
30         }
31         h1 {
32             color: #333;
33         }
34         p {
35             color: #666;
36         }
37         a {
38             display: inline-block;
39             margin-top: 20px;
40             text-decoration: none;
41             color: #0078FF;
42             font-weight: bold;
43         }
44         a:hover {
45             text-decoration: underline;
46         }
47     </style>
48 </head>
49 <body>
50     <div class="container">
51         <h1><?php echo $pageTitle; ?></h1>
52         <p><?php echo $aboutContent; ?></p>
53         <a href="index.php">Back to Home</a>
54     </div>
55 </body>
56 </html>
57
```

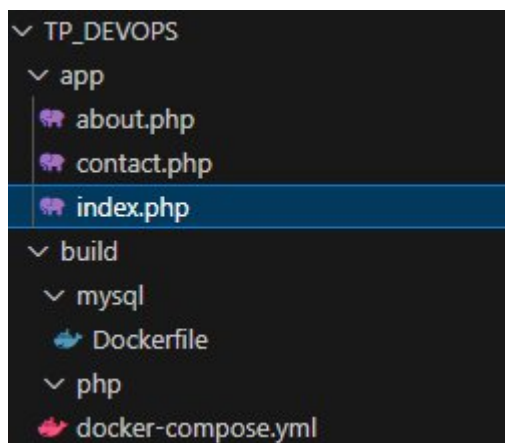
Générer Contact.php

Vim contact.php

Contenu de contact.php

```
1 <?php
2 // This is a Contact page for the website
3 $pageTitle = "Contact Us";
4 ?>
5
6 <!DOCTYPE html>
7 <html lang="en">
8 <head>
9     <meta charset="UTF-8">
10     <meta name="viewport" content="width=device-width, initial-scale=1.0">
11     <title><?php echo $pageTitle; ?></title>
12 </head>
13 <body>
14     <div class="container">
15         <h1>
16             <?php echo $pageTitle; ?>
17         </h1>
18         <form action="submit_contact.php" method="post">
19             <div>
20                 <input type="text" id="name" name="name" required>
21             </div>
22             <div>
23                 <input type="email" id="email" name="email" required>
24             </div>
25             <div>
26                 <input type="text" id="message" name="message" required>
27             </div>
28             <div>
29                 <input type="submit" value="Send Message">
30             </div>
31         </form>
32         <a href="index.php">Back to Home</a>
33     </div>
34 </body>
35 </html>
```

Configuration de l'architecture pour l'application web:



Génération du Dockerfile pour MySql:

```
1 FROM mysql:latest
2 USER root
3 RUN chmod 755 /var/lib/mysql
4
```

Génération du docker-compose.yml

```
1 version: "3.9"
2 services:
3   php-apache:
4     ports:
5       - "80:80"
6     build: '/home/marc/TP_DEVOPS/build/php'
7     volumes:
8       - /home/marc/TP_DEVOPS/app:/var/www/html
9   mysql:
10    ports:
11      - "3306:3306"
12    build: '/home/marc/TP_DEVOPS/build/mysql'
13    environment:
14      MYSQL_ROOT_PASSWORD: "Bonjour"
15      MYSQL_DATABASE: "Mon_siteweb"
16    volumes:
17      - dbData:/var/lib/mysql
18 volumes:
19   app:
20   dbData:
```

Se placer dans ~/TP_DEVOPS/build/

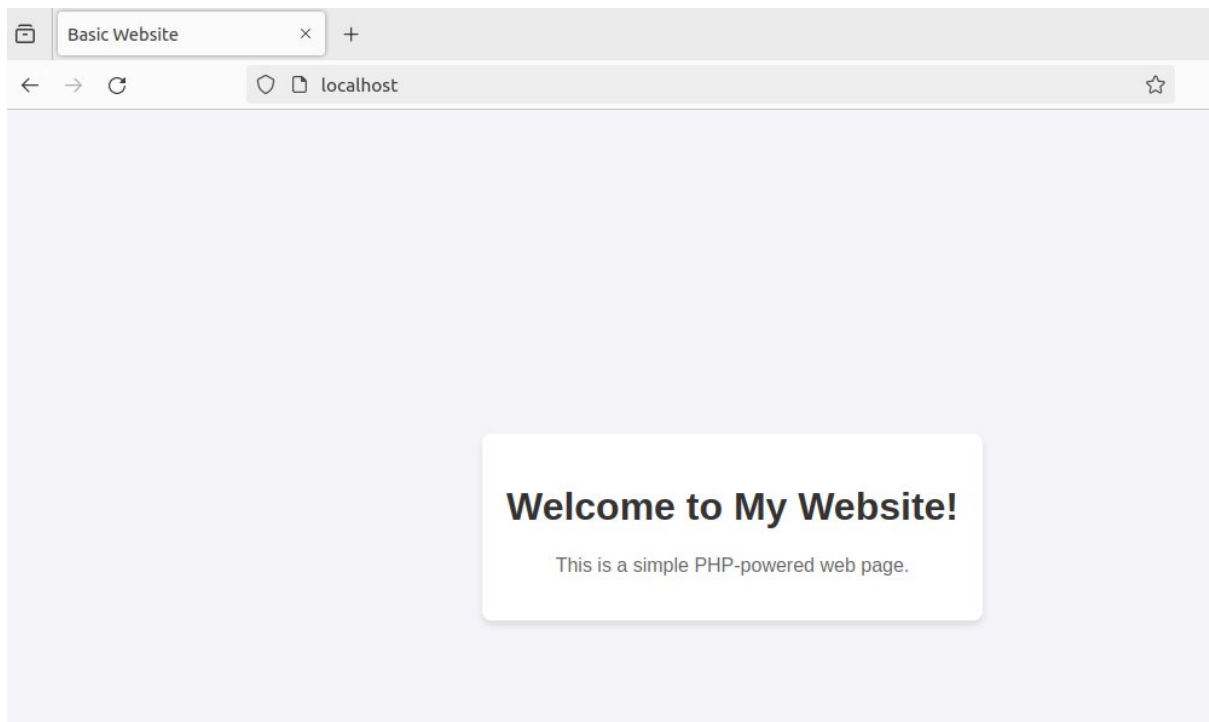
Executer la commande sudo docker-compose up

```
marc@marc-virtual-machine:~/TP_DEVOPS/build$ sudo docker-compose up
[sudo] password for marc:
Recreating build_php-apache_1 ... done
Recreating build_mysql_1 ... done
Attaching to build_mysql_1, build_php-apache_1
mysql_1 | 2024-12-19 14:20:28+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.1.0-1.el9 started.
php-apache_1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.19.0.2. Set the 'ServerName' directive globally to suppress this message
php-apache_1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.19.0.2. Set the 'ServerName' directive globally to suppress this message
php-apache_1 | [Thu Dec 19 14:20:29.113351 2024] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.62 (Debian) PHP/8.1.31 configured -- resuming normal operations
php-apache_1 | [Thu Dec 19 14:20:29.113296 2024] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
mysql_1 | 2024-12-19 14:20:29+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysql_1 | 2024-12-19 14:20:29+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.1.0-1.el9 started.
mysql_1 | '/var/lib/mysql/mysql.sock' -> '/var/run/mysql/mysql.sock'
mysql_1 | 2024-12-19T14:20:29.589453Z 0 [System] [MY-015015] [Server] MySQL Server - start.
```

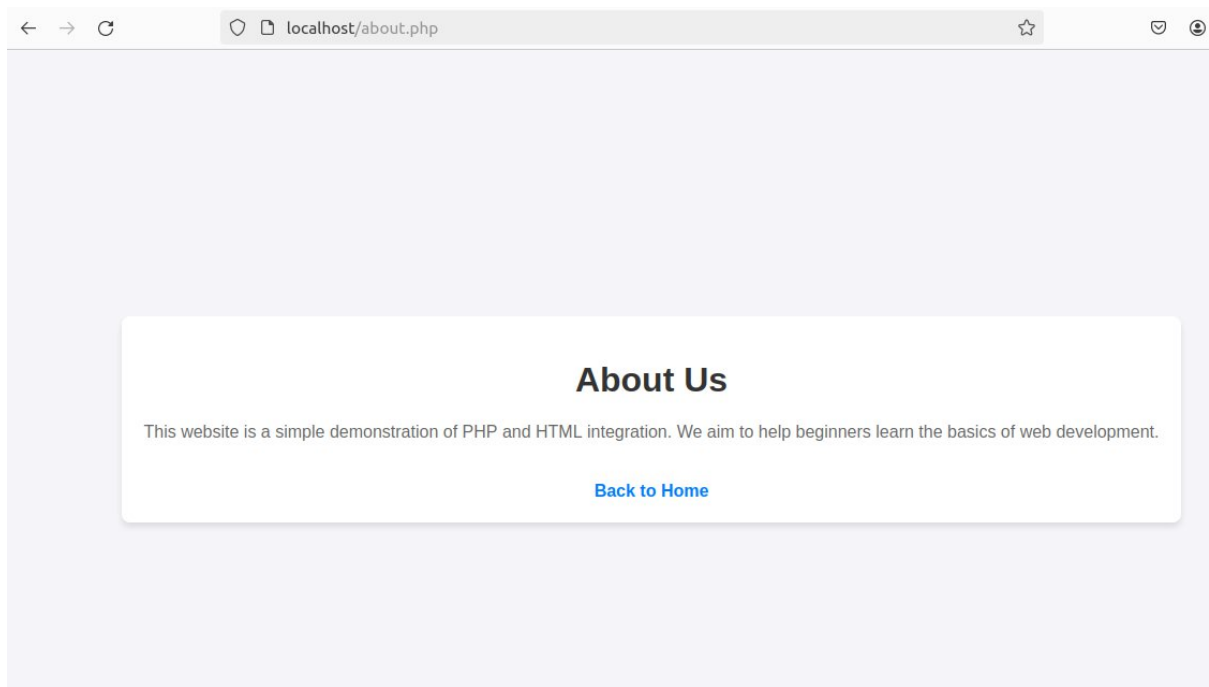
Vérification des images avec la commande sudo docker images

```
marc@marc-virtual-machine:~$ sudo docker images
[sudo] password for marc:
REPOSITORY          TAG          IMAGE ID      CREATED      SIZE
build_mysql         latest      2699ac47e8f0  2 days ago  603MB
build_php-apache    latest      9036bba349c2  2 days ago  520MB
php                 8.1-apache 173bd614b4e4  4 weeks ago  500MB
mysql              latest      56a8c14e1404  2 months ago 603MB
gcr.io/k8s-minikube/kicbase v0.0.45    aeed0e1d4642  3 months ago 1.28GB
```

Controle du fonctionnement des pages web :



About.php



Contact.php

A screenshot of a web browser window. The address bar shows 'localhost/contact.php'. The page has a light purple background. In the center, there is a white rounded rectangle containing the title 'Contact Us' in bold black text. Below the title, there are three input fields: 'Your Name:', 'Your Email:', and 'Your Message:'. Each field has a corresponding text input box. Below the 'Your Message:' field is a blue button labeled 'Send Message'. At the bottom of the white box is a blue link labeled 'Back to Home'.

